



University of Engineering & Management, Kolkata
University of Engineering & Management, Jaipur
Institute of Engineering & Management, Kolkata

B. Tech in Computer Science and Engineering
(Artificial Intelligence & Machine Learning)

Admission Year: 2024

COURSE STRUCTURE

Semester IV (Second year) Curriculum								
Sl. No	Type of course	Course Code	Course Name	Hours per week				Credit Points
				Lecture	Tutorial	Practical	Sessional	
Theory Papers								
1	Professional Core Course	PCCCS401	Discrete Mathematics	3	0	0	0	3
2	Professional Core Course	PCCCS402	Computer Organization & Architecture	3	0	0	0	3
3	Professional Core Course	PCCCS403	Artificial Intelligence & Machine Learning	3	0	0	0	3
4	Professional Core Course	PCCCS404	Design & Analysis of Algorithms	3	0	0	0	3
5	Professional Core Course	PCCCS405	Advanced Programming (OOP)	3	0	0	0	3
6	Humanities & Social Sciences including Management course	HSMCS471	Management - 1 (Finance & Accounting)	3	0	0	0	3
7	Humanities & Social Sciences including Management course	ESP401	Essential Studies for Professionals - IV	2	0	0	0	0.5
8	Mandatory Course	MCC471	Sustainability, Climate Actions & Environmental Sciences	1	-	- (Field Projects)	-	2

	Total			21	0	0	0	20.5
Practical Papers								
1	Professional Core Course	PCCCS492	Computer Organization & Architecture Laboratory	0	0	4	0	2
2	Professional Core Course	PCCCS493	Artificial Intelligence & Machine Learning Laboratory	0	0	2	0	1
3	Professional Core Course	PCCCS494	Design & Analysis of Algorithms Laboratory	0	0	4	0	2
4	Professional Core Course	PCCCS495	Advanced Programming (OOP) Laboratory	0	0	2	0	1
	Total			0	0	12	0	6
Sessional Papers								
1	Humanities & Social Sciences including Management course	SDP481	Skill Development for Professionals - IV	0	0	0	2	0.5
2	Innovative Project	PRJCS481	Innovative Project – II	0	0	0	0	1
3	Professional Core Course	PCCCS481	Data Analytics	0	0	0	2	1
	Total			0	0	0	4	2.5
Mandatory Requirements								
Sl. No	Type of course	Course Code	Course Name	Hours per week				Score/Credit /Count

1	Co-curricular & Extra Curricular Activities	MAR	Mandatory Additional Requirement s (Score)	-	-	-	-	-
2	Honours	MOOCs	Massive Open Online Course (Credit)	-	-	-	-	-
3	Certification	IFC	Industry and Foreign Certification (Count)	-	-	-	-	-
4			Skill Activity Report	-	-	-	-	-
Total				21	0	12	4	29



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Department of Computer Science and Engineering
(Artificial Intelligence & Machine Learning)

B.Tech in Computer Science and Engineering (Artificial Intelligence & Machine Learning)
DETAILED SYLLABUS

Course Code- PCCCS401

Course Title – Discrete Mathematics

Credit – 3

Category – Professional Core Course

Semester – IV

L:T:P:S – 3:0:0:0

Study material	Coursera	Linkedin	NPTEL
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Module No.	Topic	Sub-topics	Mapping with Industry and International Academia	Lecture Hours	Corresponding Lab Assignments	Textbook Mapping
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1	Sets, Relation and Function, Principles of mathematical induction	Operations and Laws of Sets, Cartesian Products, Binary Relation, Partial Ordering Relation, Equivalence Relation, Image of a Set, Sum and Product of Functions, Bijective functions, Inverse and Composite Function, Size of a Set, The Well-Ordering Principle, The Division algorithm, Prime Numbers, The Greatest Common Divisor, Euclidean Algorithm, The Fundamental Theorem of Arithmetic. Coprimality (or Euler's totient function), Chinese Remainder Theorem.	International Academia: https://ocw.mit.edu/courses/18-310-principles-of-discrete-applied-mathematics-fall-2013/pages/syllabus/ AICTE-prescribed syllabus: https://www.aicte-india.org/sites/default/files/Model_Curriculum/AICTE%20-%20UG%20CSE.pdf Industry Mapping: https://www.sagemath.org/MATLAB	10	Implement Euclidean algorithm using C/Python; Implement RSA algorithm using C/Python; Implement Fermat's little theorem / Primality checking using C / Python; Check if any two given number is co-prime using Python / C.	Discrete Mathematics and Application by Kenneth Rosen, 8th Edition. Chapters: 2, 4, 5, 9
2	Basic counting techniques, Propositional Logic, Proof Techniques	Basic counting techniques: Inclusion and exclusion principle, pigeon-hole principle, permutation and combination; recurrence relations, generating functions. Propositional Logic: Syntax, Semantics, Validity and Satisfiability, Basic	International Standards: https://ocw.mit.edu/courses/18-310-principles-of-discrete-applied-mathematics-fall-2013/pages/syllabus/ AICTE prescribed syllabus:	14	Constructing n-SAT/3-SAT solver using C/Python; Constructing propositional logic examples using Python;	Discrete Mathematics and Application by Kenneth Rosen, 8th Edition. Chapters: 1, 6, 8, 12

		Connectives and Truth Tables, Logical Equivalence: The Laws of Logic, Logical Implication, Rules of Inference, The use of Quantifiers. Proof Techniques: Some Terminology, Proof Methods and Strategies, Forward Proof, Proof by Contradiction, Proof by Contraposition, Proof of Necessity and Sufficiency. Boolean Algebra: Identities of Boolean Algebra, Duality, Representation of Boolean Function, Disjunctive and Conjunctive Normal Form.	https://www.aicte-india.org/sites/default/files/Model_Curriculum/AICTE%20-%20UG%20CSE.pdf Industry Mapping:			
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3	Algebraic Structures and Morphism	Algebraic Structures with one Binary Operation, Semi Groups, Monoids, Groups, Permutation Groups, Normal Subgroups, Ring, Field, Vector spaces, Inner-product spaces	International Standards: https://ocw.mit.edu/courses/18-310-principles-of-discrete-applied-mathematics-fall-2013/pages/syllabus/ AICTE prescribed syllabus: https://www.aicte-india.org/sites/default/files/Model_Curriculum/AICTE%20-%20UG%20CSE.pdf Industry Mapping: https://www.sagemath.org/MATLAB	6	Conversion of First Order Logic statements to Conjunctive Normal Form using Python/ SAGEMATH; Conversion of First Order Logic statements to Disjunctive Normal Form using Python/ SAGEMATH;	A BOOK OF ABSTRACT ALGEBRA by Charles C. Pinter Chapters: 1, 2, 3, 5, 7, 8, 17, 28
4	Graphs and Trees	Graphs and their properties, Degree, Connectivity, Path, Cycle, Sub Graph, Isomorphism, Eulerian and Hamiltonian Walks, Graph Coloring, Planar Graphs, Matching, Trees	International Standards : https://ocw.mit.edu/courses/18-310-principles-of-discrete-applied-mathematics-fall-2013/pages/syllabus/ AICTE prescribed syllabus:	6	Implementation of maximum flow problem using Python; Checking a graph is Hamiltonian using Python / SAGEMATH.	Discrete Mathematics and Application by Kenneth Rosen, 8th Edition. Chapter: 10

			https://www.aicte-india.org/sites/default/files/Model_Curriculum/AICTE%20-%20UG%20CSE.pdf Industry Mapping: https://www.sagemath.org/MATLAB			
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Textbooks:

1. Discrete Mathematics and Application by Kenneth Rosen, 8th Edition
2. A BOOK OF ABSTRACT ALGEBRA by Charles C. Pinter, 2nd Edition

Reference books:

1. Introductory Discrete Mathematics by V. K. Balakrishnan, Prentice Hall

Course Code- PCCCS402

Course Title – Computer Organization & Architecture

Credit – 3

Category – Professional Core Course

Semester – IV

L:T:P:S– 3:0:0:0

Pre-requisite – Digital Electronics, Basic Electronics

Course Outcomes:

CO1	Ability to design the Instruction set Architecture of a Computer System
CO2	Ability to design Arithmetic Logic Unit and Control Unit to execute the instructions
CO3	Ability to apply the concepts in Memory Organization
CO4	Ability to understand parallel processing

Relevant Links:

<u>Study Material</u>	<u>Coursera</u>	<u>NPTEL</u>	<u>LinkedIn Learning</u>	<u>Infosys Springboard</u>
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Module No.	Topic	Sub-topics	Mapping with Industry and International Academia	Lecture Hours	Corresponding Lab Assignments	Textbook Mapping
1	Introduction and Performance Evaluation	Role of abstraction, basic functional units of a computer, Stored Program Architecture, Von-Neumann model of computation, A note on Moore's law, clocking methodology, Amdahl's law, Notion of IPC, and performance.	International Academia: https://ocw.mit.edu/courses/6-823-computer-system-architecture-fall-2005/pages/lecture-notes/ AICTE-prescribed syllabus: https://www.aicte-india.org/sites/default/files/Model_Curriculum/Updated-AICTE%20-%20UG%20CSE.pdf Industry Mapping: SPEC (https://www.spec.org)	3	1. Familiarization of Hardware assembling for a digital computer. 2. Familiarization of SPEC Benchmark Application for CPU.	Computer Organization and Design: The Hardware/Software Interface”, David A. Patterson and John L. Hennessy, 5th Edition, Elsevier. Chapter: 1

2	Data representation and basic operations	Fixed and floating point (IEEE 754 Single and double precision format) representation of numbers; Overflow; Design of Adders - Ripple Carry Adder, Carry Look Ahead Adder, multiplication - shift-and-add, Booth multiplier, carry save multiplier, etc. Division - non-restoring and restoring techniques, floating point arithmetic.	International Academia: https://web.stanford.edu/class/cs107/ AICTE-prescribed syllabus: https://www.aicte-india.org/sites/default/files/Model_Curriculum/Updated-AICTE%20-%20UG%20CSE.pdf Industry Mapping: VHDL - online platform (https://www.edapla.com/) VHDL- Xilinx ISE Hardware Chipsets (TTL IC Chipsets 7400, 7402, 7404, 7408, 7432, 7486, 74151, 74153, 7483; CMOS IC Chipsets	7	1. Implementation of Half Adder, Half Subtractor, Full Adder, Full Subtractor using VHDL (Dataflow Model). 2.a) Implementation of Full Adder using VHDL (Behavioral Model). 2.b) Implementation of n-bit Carry propagation adder in VHDL (Behavioral Model). 3. Implementation of 4:1 MUX using 2:1 MUX (using Structural Method) in VHDL. 4. Implementation of signed	1. Computer System Architecture: Third Edition, Morris Mano. Chapter: 3 2. Computer Organization : Fifth Edition Carl Hamacher, Zvonko Vranesic and Safwat Zaky. Chapter: 6
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					multiplier using VHDL. 5. Implementation of signed multiplier using VHDL. 6. Implementation of Non-Restoring Division algorithm using VHDL. 7. Realization of Boolean Expressions Using Basic Gates (IC Chips). 8. Design an 8 to 1 multiplexer unit (MUX) using basic gates and using IC 74151. 9. Design of A 4-Bit Parallel Binary Adder Circuit Using The IC-Chip 7483. 10. Use a	
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					multiplexer unit to design a composite ALU [ALU Logic circuit, shift circuit and arithmetic circuit].11. Implementation of Full adder using FPGA kit.	
3	Instruction Set Architecture	CPU registers, instruction format and encoding, addressing modes, instruction set, instruction types, instruction decoding and execution, basic instruction cycle, Reduced Instruction Set Computer (RISC), Complex Instruction Set Computer (CISC), Case study - instruction sets of some common CPUs.	International Academia: https://ocw.mit.edu/courses/6-823-computer-system-architecture-fall-2005 https://www.cse.iitd.ac.in/~srsarangi AICTE-prescribed syllabus: https://www.aicte-india.org/sites/default/files/Model_Curriculum/Updated-AICTE%20-%20UG%20CSE.pdf Industry Mapping: Keil MDK https://www.keil.com	6	1. Generate Happy numbers. 2. Generate Autonomic numbers 3. Generate Hardy-Ramanujan number 4. Implement a 4-function calculator.	Computer System Architecture: Third Edition, Morris Mano. Chapter: 8

			<u>TRACE32 Simulator (https://www.lauterbach.com)</u> <u>Arm Instruction Emulator (https://developer.arm.com/Tools%20and%20Software/Arm%20Instruction%20Emulator)</u>			
4	Processor Design	Hardwired and micro-programmed design approaches, Case study - design of a simple hypothetical CPU	International Academia: <u>https://ocw.mit.edu/courses/6-823-computer-system-architecture-fall-2005</u> AICTE-prescribed syllabus: <u>https://www.aicte-india.org/sites/default/files/Model_Curriculum/Updated-AICTE%20-%20UG%20CSE.pdf</u> Industry Mapping: <u>Keil MDK (https://www.keil.co)</u>	3	Design a primitive CPU for the given instruction subset- i) data transfer ii) arithmetic operations iii) logical operations iv) branch statements	1. Computer System Architecture: Third Edition, Morris Mano. Chapter: 7 2. Computer Organization : Fifth Edition Carl Hamacher, Zvonko Vranesic and Safwat Zaky.

			m TRACE32 Simulator https://www.lauterbach.com Arm Instruction Emulator https://developer.arm.com/Tools%20and%20Software/Arm%20Instruction%20Emulator			Chapter: 7
5	Memory hierarchy	Memory hierarchy; Main memory organization - paging, segmentation, virtual memory; Cache memory- different indexing mechanisms, Trade-offs related to block size, associativity, and cache size, Processor-cache interactions for a read/write request, basic optimizations like write through/write-back caches, Average memory access time, Cache replacement policies (LRU), locality of reference, Memory interleaving; introduction to magnetic disks (notion of tracks, sectors).	International Academia: https://ocw.mit.edu/courses/6-823-computer-system-architecture-fall-2005 AICTE-prescribed syllabus: https://www.aicte-india.org/sites/default/files/Model_Curriculum/Updated-AICTE%20-%20UG%20CSE.pdf Industry Mapping: VHDL - online platform (https://www.edapla	7	1. Implementation of memory unit consisting of 16X4 RAM and 8X 4 ROM. 2. Implement Read Write operation using 16X4 RAM.	Computer System Architecture: Third Edition, Morris Mano. Chapter: 12

			yground.com/ VHDL- Xilinx ISE			
6	Input/Output Organization	Programmed I/O, Interrupt-driven I/O, and DMA	International Academia: https://ocw.mit.edu/courses/6-823-computer-system-architecture-fall-2005 AICTE-prescribed syllabus: https://www.aicte-india.org/sites/default/files/Model_Curriculum/Updated-AICTE%20-%20UG%20CSE.pdf Industry Mapping: ATMEGA 16 Microcontroller	2	1. Interface LCD with ATmega16 and display your name continuously. 2. Interface LCD with ATmega16 to display any string with a blinking cursor at the end.	Computer System Architecture: Third Edition, Morris Mano. Chapter: 11
7	Parallel Processing	Pipelining - Basic concepts, instruction and arithmetic pipeline, different types of dependencies and hazards, techniques for handling hazards, Pipeline optimization techniques - reservation table; Superscalar, super pipelined and VLIW	International Academia: https://ocw.mit.edu/courses/6-823-computer-system-architecture-fall-2005 AICTE-prescribed syllabus: https://www.aicte-india.org/sites/default/files/Model_Curriculum/Updated-AICTE%20-%20UG%20CSE.pdf	6	1) Implement a Pipelined Multiplier using VHDL and FPGA Kit. 2) Implement a Pipeline Control Unit using RIPS	“Computer Organization and Design: The Hardware/Software Interface”, David A. Patterson and John L. Hennessy,

		processor architectures; Array and vector processors; Multiprocessor architecture: taxonomy of parallel architectures; Centralized shared-memory architecture and distributed shared memory architecture; Cache Coherence;	india.org/sites/default/files/Model_Curriculum/Update_d-AICTE%20-%20UG%20CSE.pdf Industry Mapping: RIPES https://github.com/mortbopet/Ripes VHDL - online platform https://www.edaplayground.com, VHDL- Xilinx ISE			5th Edition, Elsevier. Chapters: 4 and 6
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Textbooks:

1. Computer System Architecture: Third Edition, Morris Mano.
2. Computer Organization: Fifth Edition Carl Hamacher, Zvonko Vranesic and Safwat Zaky.
3. Computer Organization and Design: The Hardware/Software Interface: David A. Patterson and John L. Hennessy.

Reference books:

1. Computer Organization and Architecture – Designing for Performance: William Stallings
2. Computer Architecture and Organization: John P Hayes
3. Computer Architecture and Parallel Processing: K. Hwang, F. A. Briggs

List of Assignments:

- 1) Familiarization of Hardware assembling for a digital computer.
- 2) Familiarization of SPEC Benchmark Application for CPU.
- 3) Realization of Boolean Expressions Using Basic Gates (IC Chips).
- 4) Design an 8 to 1 multiplexer unit (MUX) using basic gates and using IC 74151.
- 5) Design of A 4-Bit Parallel Binary Adder Circuit Using The IC-Chip7483.
- 6) Implementation of Half Adder, Half Subtractor, Full Adder, Full Subtractor using VHDL and Verilog (Dataflow Model).
- 7) a) Implementation of Full Adder using Verilog (Behavioral Model). b) Implementation of n-bit Carry propagation adder in VHDL (Behavioral Model).
- 8) Implementation of 4:1 MUX using 2:1 MUX (using Structural Method) in VHDL.
- 9) Implementation of signed multiplier using VHDL.
- 10) Implementation of Non-Restoring Division algorithm using VHDL.
- 11) Use a multiplexer unit to design a composite ALU [ALU Logic circuit, shift circuit and arithmetic circuit].
- 12) Implementation of Full adder using FPGA kit.
- 13) Design a primitive CPU for the given instruction subset-
 - i) data transfer
 - ii) arithmetic operations
 - iii) logical operations
 - iv) branch statements
14. Interface LCD with ATmega16 and display your name continuously.
15. Interface LCD with ATmega16 to display any string with a blinking cursor at the end.
16. Implementation of memory unit consisting of 16X4 RAM and 8X 4 ROM.
17. Implement Read Write operation using 16X4 RAM.
18. Implement a Pipelined Multiplier using VHDL and FPGA Kit.

CO-PO Mapping:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	3	3	2	2	2	3	2	2	3
CO2	3	3	3	3	3	2	2	2	3	2	2	3
CO3	3	3	3	3	3	2	2	2	3	2	2	3
CO4	3	3	2	3	2	2	2	2	3	2	2	3

Course Code- PCCCS403

Course Title – Artificial Intelligence & Machine Learning

Credit – 3

Category – Professional Core Course

Semester – IV

L:T:P – 3:0:0

Module No.	Topic	Sub-topics	Mapping with Industry and International Academia	Lecture Hours	Corresponding Lab Assignments	Textbook Mapping
1	Introduction to Artificial Intelligence	Definition and Scope of AI - History & Evolution of AI, Sub fields of AI, Ethical Considerations Problem Solving in AI – Searching, Optimization, Planning, Game Theory	International Academia: https://ocw.mit.edu/courses/6-867-machine-learning-fall-2006/pages/syllabus/ AICTE-prescribed syllabus: https://www.aicte-india.org/sites/default/files/Model_Curriculum/AICTE%20-%20UG%20CSE.pdf Industry Mapping: <i>AI using Python</i>	5	1. Basic Practice programs using Python. 2. Introduction to Weka Toolkit.	

		Knowledge Representation – Propositional Logic, Predicate Logic				
2	Problem solving techniques	State space search, control strategies, heuristic search, problem characteristics, production system characteristics., Generate and test, Hill climbing, best first search, A* search, Constraint satisfaction problem, Mean- end analysis, Min-Max	AICTE-prescribed syllabus: https://www.aicte-india.org/sites/default/files/UG_Emerging.pdf International Academia: https://ocw.mit.edu/courses/6-034-artificial-intelligence-fall-2010/resources/lecture-1-introduction-and-scope/ Industry Mapping: Python		1. Python programming, symbolic algebra, Water Jug Problem 2. Perform BFS and DFS analysis on Facebook SN Dataset 3. Implementation of A* search 4. N-Queens Problem	

		Search, Alpha-Beta Pruning, Additional refinements, Iterative Deepening.				
3	Introduction to Machine Learning	Machine Learning Definition & Scope – Evolution of Learning Systems Data Representation – Features, Covariance, Orthogonality, Feature Selection vs Feature Extraction overview, Missing Data, Outliers, Noise, Normalization, Standardization,	<i>International Academia: https://ocw.mit.edu/courses/6-867-machine-learning-fall-2006/pages/syllabus/ AICTE-prescribed syllabus: https://www.aicte-india.org/sites/default/files/Model_Curriculum/AICTE%20-%20UG%20CSE.pdf Industry Mapping: <i>Python, Weka Toolkit</i></i>		1. Load and Interpret Data with Weka in the arff format or csv format using Python. 2. Distinguish between normalization and standardization of datasets. 3. Implement an algorithm in Python to handle missing data.	

		Training Validation and Testing. Model Selection- Holdout, Cross validation, k- fold Cross validation, random sampling, Bias- Variance tradeoff, overfitting, underfitting. Machine Learning Types – Supervised, Unsupervised, Reinforcement, Adversarial, Meta- Learning Zero-shot & One-shot Learning, Transfer			4. Data visualization using- Histogram Plot, Scatter Plot etc.	
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		Learning, Ensemble Learning – Bagging and Boosting.				
4	Supervised Learning	Regression – Linear Regression, Polynomial Regression, regularization Classification – Logistic Regression, Decision Trees, Bayesian Classifier, Parametric and Non-parametric estimation of probability densities (Maximum Likelihood Estimate, Bayesian Estimation),	International Academia: https://ocw.mit.edu/courses/6-867-machine-learning-fall-2006/pages/syllabus/ https://ocw.mit.edu/courses/6-046j-design-and-analysis-of-algorithms-spring-2015/pages/syllabus/ AICTE-prescribed syllabus: https://www.aicte-india.org/sites/default/files/Model_Curriculum/AICTE%20-%20UG%20CSE.pdf Industry Mapping: Python and Weka Toolkit, Sci-kit Learning and Matplotlib Visualization		1. Using Weka and Python compare the performance of Decision Trees, Random Forest, Naïve Bayes and K-Nearest Neighbor algorithms. 2. Using LibSVM implement support vector machines and compare the different kernel	

		Naïve Bayes Classifier, K- Nearest Neighbours, Support Vector Machines Representation Learning – Perceptron, activation functions, XOR problem, Multi-layered Perceptron, Gradient Descent, Backpropagation Performance Metrics – Confusion Matrix, Precision, Recall, F-Measure, Area under ROC Curve.		functions. 3. Implement K Nearest Neighbor Algorithm 4. Implement a perceptron using python and develop the perceptron training rule. Project 1 - Write a code in python to create a multi-layered perceptron which can be trained using backpropagation using Numpy and	
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					visualize the training using Matplotlib	
5	Unsupervised Learning	Dimensionality Reduction : Eigen Value Decomposition, Principal Component Analysis, Linear Discriminant Analysis, Singular Value Decomposition Clustering : Gaussian Mixture Models, Expectation Maximization, K-	International Academia: https://ocw.mit.edu/courses/6-867-machine-learning-fall-2006/pages/syllabus/ AICTE-prescribed syllabus: https://www.aicte-india.org/sites/default/files/Model_Curriculum/AICTE%20-%20UG%20CSE.pdf Industry Mapping: Weka Toolkit , Sci-kit Learning and Matplotlib Visualization		1. Use Weka and Python to implement various feature extraction techniques 2. Visualize data clusters using t-SNE visualization and silhouette coefficient 3. Implement K-Means Clustering 4. Find the correct value of K in K Means Clustering using Cluster Validity Indices. Project 2:	

		Means Algorithms , Hierarchical Clustering (AGNES) Metrics: Similarity Measures, silhouette coefficient, Cluster Linkage Metrics, Cluster Validity Indices.			Compare various clustering algorithms to cluster aerial images and visualize the results.	
6	Reinforce ment Learning	Definition and Key Concepts: Agent, Environment , Reward, Policy, Value Function Algorithms : Q-Learning, Exploration vs Exploitation	International Academia: https://ocw.mit.edu/courses/6-867-machine-learning-fall-2006/pages/syllabus/ https://ocw.mit.edu/courses/6-046j-design-and-analysis-of-algorithms-spring-2015/pages/syllabus/ AICTE-prescribed syllabus: https://www.aicte-india.org/sites/default/files/Model_Curriculum/AICTE%20-%20UG%20CSE.pdf Industry Mapping: Open AI Gymnasium		Project 3: Use the OpenAI Gymnasium to train a Reinforcement Learning Bot.	

		, Convergence, Issues and Challenges				
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Textbooks:

1. Machine Learning by Tom Mitchell. – Mc Graw Hill
2. Machine Learning by S. Sridhar, M. Vijayalakshmi – Oxford University Press
3. Machine Learning: The Art and Science of Algorithms That Make Sense Of Data – Peter Flach, Cambridge
4. Pattern Recognition and Machine Learning by Christopher Bishop - Springer NP Exclusive (CBS)
5. Machine Learning: Theory and Practice - M.N. Murty, V.S. Ananthanarayana, Universities Press
6. Artificial Intelligence: A Modern Approach - Russell, Norvig, Pearson
7. Artificial Intelligence - George Luger, Pearson (Indian Edition)
8. First Course In Artificial Intelligence - Deepak Khemani, McGraw-Hill
9. An Introduction to Statistical Learning - James, Witten, Hastie, Tibshirani, Springer

Course Code- PCCCS404

Course Title – Design & Analysis of Algorithm

Credit – 3

Category – Professional Core Course

Semester – IV

L:T:P:S – 3:0:0:0

Module No.	Topic	Sub-topics	Mapping with Industry and International Academia	Lecture Hours	Corresponding Lab Assignments	Textbook Mapping
1	Introduction	Characteristics of algorithm. Analysis	International Academia: https://ocw.mit.edu/courses/6-046j-design-and-analysis-of-algorithms-spring-2015/pages/syllabus/ AICTE-prescribed syllabus:	5	1. Verify the different input case	

		of algorithm: Asymptotic analysis of complexity bounds – best, average and worst-case behavior; Performance measurements of Algorithm, Time and space trade-offs, Analysis of recursive algorithms through recurrence relations: Substitution method, Recursion tree method and	https://www.aicte-india.org/sites/default/files/Model_Curriculum/AICTE%20-%20UG%20CSE.pdfIndustry Mapping: GNU C Compiler		scenario with insertion sort, Selection sort and Bubble sort. 2. Verify the different input case scenario with linear search and binary search.	
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		Master's theorem; Divide and Conquer algorithms – Merge Sort, Quick Sort, Finding lower bound of comparison-based sorting algorithms, Strassen's algorithm for multiplying matrices.				
2	Fundamental Algorithmic Strategies	Brute-force, Greedy, Dynamic Programming, Branch and Bound and Backtracking	International Academia: https://ocw.mit.edu/courses/6-046j-design-and-analysis-of-algorithms-spring-2015/pages/syllabus/AICTE-prescribed-syllabus : https://www.aicte-india.org/sites/default/files/Model_Curriculum/AICTE%20-%20UG%20CSE.pdf Industry Mapping: <i>Hardware Chipsets (IC 7408, 7432, 7404), Software- Tinker Cad</i>	5	1. Find maximum and Minimum using the divide & conquer approach and verify its	

		ng methodolo gies for the design of algorithms ; Illustration s of these techniques for Problem solving, Bin Packing, Knapsack, TSP, Heuristics – characteris tics and their applicatio n domains, KMP algorithm.			efficiency. 2. Verify the different input case scenario with Quick sort, Merge sort and heap sort. 3. Verify quick sort with different pivot position. 4. Verify Strassen's matrix multiplicati on.	
3	Graph and Tree Algorithms	Graph and Tree Algorithm s: Traversal algorithms : Depth	International Academia: https://ocw.mit.edu/courses/6-046j-design-and-analysis-of-algorithms-spring-2015/pages/syllabus/AICTE-prescribed-syllabus: https://www.aicte-india.org/sites/default/files/Model_Curriculum/AICTE%20-%20UG%20CSE.pdfIndustry Mapping: <i>Hardware Chipset (IC 7408, 7432,7404, 74153, 74155,</i>	6	1. Review of BFS/DFS. 2. Checking if a graph is	

		First Search (DFS) and Breadth First Search (BFS), Disjoint Set Data Structures, Shortest paths algorithms, Minimum Spanning Tree, Topological sorting, Network Flow Problem.	74180) <i>Software: LogiSim and VHDL</i>		biconnecte d. 3. Review of Spanning Tree. 4. Verify Knapsack problem. 5. Job sequence with deadline.	
4	Tractable and Intractable Problems	Tractable and Intractable Problems: Computability of Algorithms, Computability classes –	International Academia: https://ocw.mit.edu/courses/6-046j-design-and-analysis-of-algorithms-spring-2015/pages/syllabus/AICTE-prescribed-syllabus : https://www.aicte-india.org/sites/default/files/Model_Curriculum/AICTE%20-%20UG%20CSE.pdf Industry Mapping: <i>Hardware Chipset (IC 7476, 7474)</i> <i>Software: LogiSim and VHDL</i>	6		

		P, NP, NP- complete and NP- hard, Cook's theorem, Standard NP- complete problems and Reduction techniques .				
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5	Advanced Topics	Advanced Topics: Approximation algorithms , Randomized algorithms , Class of problems beyond NP – P SPACE.	International Academia: https://ocw.mit.edu/courses/6-046j-design-and-analysis-of-algorithms-spring-2015/pages/syllabus/ AICTE-prescribed syllabus: https://www.aicte-india.org/sites/default/files/Model_Curriculum/AICTE%20-%20UG%20CSE.pdf Industry Mapping: Hardware Chipset (IC 7408, 7432, 7404), FPGA Kit	4		
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Textbooks:

1. Introduction to Algorithms, 4th Edition, Thomas H Cormen, Charles E Lieserson, Ronald L Rivest and Clifford Stein, MIT Press/McGraw-Hill.
2. Algorithms In A Nutshell, George T. Heineman, Gary Pollice and Stanley Selkow, O'Reilly.
3. Fundamental pf Algorithms – E. Horowitz et al.
4. Algorithm Design, 1st Edition, Jon Kleinberg and EvaTardos, Pearson.
5. Algorithm Design: Foundations, Analysis, and Internet Examples, Second Edition, Michael T Goodrich and Roberto Tamassia, Wiley.
6. Algorithms – A Creative Approach, 3rd Edition, UdiManber, Addison-Wesley, Reading, MA.

7. Design & Analysis of Algorithms, Gajendra Sharma, Khanna Publishing Housh (AICTE Recommended Textbook – 2018).

Course Code- PCCCS405

Course Title – Advanced Programming (OOP)

Credit – 3

Category – Professional Core Course

Semester – IV

L:T:P :S – 3:0:0:0

Desirable/Advanced outcome:

1. Ability to implement basic event-driven programming.
2. Understanding of the fundamentals of parallel programming.

Mod ule No.	Topic	Sub- topics	Mapping with Industry and International Academia	Lect ure Hou rs	Corresponding Lab Assignments	Textbook Mapping
1	Familiarity with the programming environment	Understanding the build system, IDE, debugging, profiling and source code	International Academia: https://drive.google.com/file/d/10z00dMd26WjiPThhCercGbsi6u3ciE62/view https://drive.google.com/file/d/1k3qrDL9p5_IJR_iP2mt6c6AzwmByNtf/view AICTE-prescribed syllabus: https://www.aicteindia.org/sites/default/files/Model_Curriculum/AICTE%20-%20UG%20CSE.pdf Industry Mapping:	3	Familiarity with terminal/command prompt, using git commands and github to pull/ commit/ push/ merge code, writing, compiling and running simple programs,	Herbert Schildt, Java: The Complete Reference, 11th edition Chapters: 1,2

		management. Introduction to various programming paradigms, advantages of OOP, comparison of OOP with Procedural Paradigm	<i>Hackerrank, TCS Codevita projects, GitHub platform. NetBeans and Eclipse IDE will be used.</i>		debugging by setting breakpoints	E. Balaguruswami, Programming with Java, 6th edition Chapters: 1,2,3
2	Basic principles of the object-oriented development process	Introduction to Object Oriented Paradigm : Data encapsulation, modularity, code reuse, identifying classes, attributes, methods and	International Academia: https://ocw.mit.edu/courses/6-096-introduction-to-c-january-iap-2011/270def7b1f68535b7c3846c606b220eb_MIT6_096IAP11_lec07.pdf https://ocw.mit.edu/courses/6-00sc-introduction-to-computer-science-and-programming-spring-2011/resources/lecture-11-oop-and-inheritance/ AICTE-prescribed syllabus: https://www.aicteindia.org/sites/default/files/Model_Curriculum/AICTE%20-%20UG%20CSE.pdf Industry Mapping:	4	Importing pre-written classes using the this keyword, calling and defining methods, writing and instantiating classes, setter/getter methods, instance variables, returning values, debugging using	Herbert Schildt, Java: The Complete Reference, 11st edition Chapters: 3,4,5 E. Balaguruswami, Programming with Java, 6th edition

		objects, class relationships	<i>Hackerrank, TCS Codevita projects, GitHub platform. NetBeans and Eclipse IDE will be used.</i>		print function, containment and association, scope and parameter passing	Chapters: 6,7,8
3	Advanced features of OOP	Polymorphism, Inheritance, abstract classes & Interfaces, copying and cloning objects, Wrapper class, Stream, I/O operations	International Academia: <i>International Standards Mapping (MIT Open Courseware): String: immutability, BufferedReader, StringBuilder</i> <u>https://ocw.mit.edu/courses/6-088-introduction-to-c-memory-management-and-c-object-oriented-programming-january-iap-2010/resources/mit6_088iap10_lec05/</u> <i>Polymorphism: Methods: Overloading and overriding Methods, overloading constructors</i> <u>https://ocw.mit.edu/courses/6-092-introduction-to-programming-in-java-january-iap-2010/download/</u> AICTE-prescribed syllabus: <u>https://www.aicteindia.org/sites/default/files/Model_Curriculum/AICTE%20-%20UG%20CSE.pdf</u> Industry Mapping: <i>Hackerrank, TCS Codevita projects, GitHub platform. NetBeans and Eclipse IDE will be used.</i>	8	Parameter polymorphism, method resolution, declared v/s actual type, partially and fully overriding methods, calling superclass constructor from child class constructor, protected fields and methods, using an abstract parent class v/s an interface with default and abstract methods, object equality check, object comparison (Comparable/Comparator interface),	Herbert Schildt, Java: The Complete Reference, 11th edition Chapters: 7,8,20 E. Balaguru swami, Programming with Java, 6th edition Chapters: 9,10

					Cloneable interface/copy constructor	
4	Access modifiers, Exceptions & Multithreading	Access modifiers, packages, Java APIs, Exception handling, Java Thread Programming	International Academia: https://drive.google.com/file/d/1yFNf2IBXgy6ch47hR6TGHzddvCPfVm8p/view AICTE-prescribed syllabus https://www.aicteindia.org/sites/default/files/Model_Curriculum/AICTE%20-%20UG%20CSE.pdf Industry Mapping: Hackerrank, TCS Codevita projects, GitHub platform. NetBeans and Eclipse IDE will be used.	6	Exception handling using try/catch block, nesting try/catch blocks, throw and throws keywords, rethrowing exceptions, handling checked exception, user defined exceptions. Thread Synchronization and Thread Communication	Herbert Schildt, Java: The Complete Reference, 11th edition Chapters: 9, 10 E. Balaguru swami, Programming with Java, 6th edition Chapters: 11, 12, 13
5	Collection Framework	Language supported libraries for handling advanced data structures (Hierarchy)	International Academia: https://drive.google.com/file/d/1kbRGF396sQPdQbA4w-N81ElKU_bdGgFs/view AICTE prescribed syllabus: https://www.aicteindia.org/sites/default/files/Model_Curriculum/AICTE%20-%20UG%20CSE.pdf Industry Mapping:	4	Time complexity analysis, Java collection framework (or Boost libraries), sorting objects, iterating	Herbert Schildt, Java: The Complete Reference, 11th edition Chapters: 18, 19

		y of Collection Framework, ArrayList, LinkedList, Vector, HashSet)	<i>Hackerrank, TCS Codevita projects, GitHub platform. NetBeans and Eclipse IDE will be used.</i>		over objects	E. Balaguru swami, Programming with Java, 6th edition Chapters: 17, 18
6	Modeling and Design patterns	Basic modeling techniques – e.g. Class diagram, sequence diagram, use case diagrams, etc. Introduction to design patterns: iterator, singleton, flyweight, adapter, strategy,	International Academia: https://drive.google.com/file/d/1DQVPfhmcyKog-PryPdZ-bxufUC7D0Cfl/view AICTE-prescribed syllabus https://www.aicteindia.org/sites/default/files/Model_Curriculum/AICTE%20-%20UG%20CSE.pdf Industry Mapping: <i>Hackerrank, TCS Codevita projects, GitHub platform. NetBeans and Eclipse IDE will be used.</i>	4	UML modeling using Rational rose, draw.io, Edraw Max, . www.starUML.io, argouml.tigris.org/	Herbert Schildt, Java: The Complete Reference, 11th edition Chapters: 18, 19 E. Balaguru swami, Programming with Java, 6th edition Chapters: 19

		template, prototype , factory, façade, decorator , composite, proxy, chain of responsib ility, observer, state) Model View Controller				
7	Basic Android Programming & UI Design	Android Componen ts – Activity, Services, Content Provider, Broadcast Receiver; Simple UI Design, Applet and Swing	International Academia: https://online.stanford.edu/courses/cs108-object-oriented-systems-design AICTE-prescribed syllabus: https://www.aicteindia.org/sites/default/files/Model_Curriculum/AICTE%20-%20UG%20CSE.pdf Industry Mapping: Hackerrank , TCS Codevita projects , GitHub platform , Android Studio will be used.	7	Android app making, stand alone and web app development.	Barry A. Burd and John Mueller, Android Application Development All-in-One For Dummies Chapters: 1,2,3,4 Rick Rogers,

						John Lombardo, Zigurd Mednieks and Blake Meike Android Application Development Chapters: 1,2,3,5
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Textbooks:

1. Herbert Schildt, Java: The Complete Reference, 11th edition
2. E. Balaguruswami, Programming with Java, 6th edition
3. Barry A. Burd and John Mueller, Android Application Development All-in-One For Dummies
4. Rick Rogers, John Lombardo, Zigurd Mednieks and Blake Meike, Android Application Development, O REILLY
5. Grady Booch, Robert A. Maksimchuk, Michael W. Engle, Bobbi J. Young, Jim Conallen, Kelli A. Houston. Object-Oriented Analysis and Design with Applications
6. Paul Deitel, Harvey Deitel, Java How to Program: Early Objects, 11e, Pearson Education

Reference books:

1. J. Rumbaugh et al. The Unified Modeling Language Reference Manual.
2. P. Van Roy and S. Haridi. Concepts, Techniques, and Models of Computer Programming.
3. Horton, I. (2005). Ivor Horton's Beginning Java 2. John Wiley & Sons.

Online Resources:

<https://missing.csail.mit.edu/>
<https://www.baeldung.com/junit>

<https://www.tutorialspoint.com/junit/index.htm>

Course Code- HSMCS471

Course Title – Management - 1 (Finance & Accounting)

Credit – 3

Category – Humanities & Social Sciences including Management Course

Semester – IV

L:T:P:S – 3:0:0:0

Module No.	Topic	Sub-topics	Mapping with Industry and International Academia	Lecture Hours	Corresponding Lab Assignments	Textbook Mapping
1	Introduction to Finance and Accounting	The principles of financial and cost accounting Financial Management, Financial Planning and Capitalization-definitions, objectives, changing roles and functions, Financial Decision. Basic accounting concepts, important definitions, uses,	International Academia: (Accounting, Finance & Valuation Course I Stanford Online) AICTE-prescribed syllabus: (Microsoft Word - Information Technology Syllabus.doc (makautwb.ac.in)) Industry Mapping: <i>Designing an accounting system</i>	8	1. Easy Tally 2. Preparation of Basic financial and accounting statements 3. Analysis of Financial statements of listed companies	

		limitations, advantages; types of Accounting, Financial statements, introduction to Journal Accounting; double entry bookkeeping, different types of transactions related to Financial Accounting.				
2	Capital Budgeting	Managerial accounting tools and practices Nature of Investment decision, Importance of Capital Budgeting, The Capital. Budgeting Process - Investment Criterion, Pay-back period, Accounting, ROR (Rate of Return) Method, Discounting Cash flow	International Academia: <u>(Accounting, Finance & Valuation Course I Stanford Online)</u> AICTE-prescribed syllabus: <u>(Microsoft Word - Information Technology Syllabus.doc (makautwb.ac.in))</u> Industry Mapping: <i>Financial long term forecasting</i>	6	Mapping and Techniques using Excel and Tally 1. Analysis of Financial statements of listed companies (International C India) comparison 2. Oracle netsuite	

		method, Net – present value method, IRR (Internal Rate of Return) method, The Benefit-Cost Ratio method. Related module			
3	Management of Working Capital	Various concepts, Elements, Classification, Financing and importance of working capital, Investment analysis, Cash flow determination, cost of capital, capital budgeting methods.	International Academia: (Accounting, Finance & Valuation Course I Stanford Online) AICTE-prescribed syllabus: (Microsoft Word - Information Technology Syllabus.doc (makautwb.ac.in)) Industry Mapping: <i>Financial daily or short term fund planning and management</i>	8	Mapping Techniques using Tally and Excel
4	Cost – Volume – Profit Analysis	Analysis of Costing and Classification of costs, Allocation, apportionment and absorption, Cost centers, different costing systems, Cost analysis for	International Academia: (Accounting, Finance & Valuation Course I Stanford Online) AICTE-prescribed syllabus: (Microsoft Word - Information Technology Syllabus.doc (makautwb.ac.in)) Industry Mapping:	8	Mapping and solving Techniques using Tally and Excel Power BI

		managerial decisions, Meaning of Linear CVP analysis, Objectives, Assumptions, Break – Even analysis, determining the Break-Even point profit, Volume graph profit, Volume ratios margin of Safety.	<i>Analysis of sales and cost dependency on profit margin using cost analysis methods.</i>			
5	Financial Control	Posting of Ledgers and preparation of Trial Balance; preparation of Balance Sheet and Profit and Loss Accounts; Controlling other departments by Financial Accounting (A practical Approach).	<i>AICTE-prescribed syllabus:</i> <i>(Microsoft Word - Information Technology Syllabus.doc (makautwb.ac.in))</i> <i>Industry Mapping:</i> <i>Design and analysis of company health using Balance sheet using available tools and techniques.</i>	6		

Course Code- MCC471

Course Title – Sustainability, Climate Actions & Environmental Sciences

Credit – 2

Category – Mandatory Course

Semester – IV

L:T:P:S – 1:0:2:0

Course Outcomes:

CO1	Understand fundamental concepts of environmental systems, sustainability, United Nations Sustainable Development Goals (UNSDGs) and their interrelationship with human society.
CO2	Apply knowledge of sustainable practices, different technical tools and existing frameworks to address environmental and societal challenges.
CO3	Analyze the challenges and strategies associated with climate change mitigation, sustainable cities, and waste management within the context of international agreements and frameworks.
CO4	Evaluate and design innovative approaches to energy, water, and waste management, considering the principles of the circular economy and global SDG progress reports.

Study Material	LinkedIn	NPTEL	Coursera
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Module No.	Topic	Sub-topics	Mapping with Industry and International Academia	Lecture Hours	Corresponding Lab Assignments	Text book Mapping
1	Overview - United Nations Sustainable	Basic ideas of environment, basic concepts: man, society &	International Academia: https://unccelearn.org/course/view.php?id=170&page=overview https://unccelearn.org/course/view.php?id=181&page=overview AICTE prescribed syllabus:	4	Assess the college campus alignment	Field work will be assigned for

	Development Goals (UNSDGs)	environment, their interrelationship. Significance of sustainability in today's world. 17 United Nations Sustainable Development Goals (UNSDGs) - background, significance, interconnectedness of goals, global challenges and recent progress. Climate change and mitigation. Explain and evaluate the evidence	Industry Mapping:		with the United Nations Sustainable Development Goals (SDGs) and rank the performance across selected goals mentioning the actionable strategies for improvement.	each and every student/group of students, on completion of which they have to give a presentation along with a model display if possible.
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		for human-caused climate change, in the context of historical climate change, as well as the relevant scientific uncertainties and possible evidence to the contrary.				
2	Sustainable Management	Sustainable management of water and sanitation-introduction, key components, challenges and innovative approaches	International Standards: https://ocw.mit.edu/courses/res-env-006-teaching-with-sustainability-january-iap-2022/ AICTE prescribed syllabus: Industry Mapping:	4	Design and propose innovative, sustainable solutions for managing water, energy, and	

		s. Ensure access to affordable, reliable, sustainable, and modern energy-introduction, importance, key targets, challenges and strategies. Sustainable Cities and Communities- Definition of sustainable cities, current challenges , strategies, innovative solution,			urban systems, inspired by the principles of SDG 6, SDG 7 and SDG 11.	
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		smart city			
3	Climat e Action	Climate change and its consequences, international agreements on climate change, strategies and actionable step, Life Below Water-Importance of water bodies and marine ecosystem, strategies for protecting aquatic life and water bodies, Life on Land-importance of biodiversity, carbon	International Standards: https://unccelearn.org/course/view.php?id=7&page=overview&lang=en https://unccelearn.org/course/view.php?id=145&page=overview https://unccelearn.org/course/view.php?id=48&page=overview https://ocw.mit.edu/courses/res-env-001-climate-action-hands-on-harnessing-science-with-communities-to-cut-carbon-january-iap-2017/ AICTE prescribed syllabus: Industry Mapping:	4	Using data analytics and modeling tools - evaluate climate change impacts, assess ecosystem health, and propose technical solutions for mitigation and

		sequestration, Food security, Strategies for Conservation and Restoration of Ecosystems , Sustainable Land Management, Biodiversity Conservation, Accountable steps for life on land. The successes and failures of past national and international efforts to address climate change, and evaluate			conservation efforts .	
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		prospects for future management of climate change. Provisions of the United Nations Framework Convention on Climate Change, Paris Agreement			
4	UN-call for Action	Focus on annual SDG Goals Report and the United Nations Secretary-General's calls for action to accelerate the progress on the Sustainable Development Goals (SDGs). Examine	International Standards: https://unccelearn.org/course/view.php?id=175&page=overview AICTE prescribed syllabus: Industry Mapping:	4	Using quantitative analysis, strategic planning, and innovative approaches, evaluate the global progression

		the global progress trends, challenges highlighted in recent reports, and key priorities proposed by the Secretary-General to achieve the 2030 Agenda.			the Sustainable Development Goals (SDGs) as highlighted in the annual SDG Goals Report.	
5	Environmental Systems Analysis	Environmental impact assessment – lifecycle assessment (LCA), Using of LCA software tools – OpenLCA, Environmental, social, and governance (ESG), Integrated	International Academia: https://www.lse.ac.uk/united-states/Assets/Documents/Syllabus-Hub-PDFs/Michael-Carbajales-Dale-Clemson-Environmental-Systems-Analysis.PDF https://www.igmpi.ac.in/environmental-social-governance?gad_source=1 https://www.iso.org/standard/60857.html#:~:text=ISO%2014001%20is%20the%20internationally,continually%20improve%20their%20environmental%20performance. Industry Mapping:	4	Life Cycle Assessment of a college building using OpenLCA software.	

		Impact Assessment of ESG, Carbon Management, Green Hydrogen, Importance of green building (LEED, IGBC etc.) certification · Environmental Management System (EMS) in industry - ISO 14001.				
6	Waste Management	Waste Management Rules - Hazardous Waste, E-waste, Municipal Solid Waste, Bio-medical waste, Plastic	International Academia: https://cpcb.nic.in/rules-6/ https://ocw.mit.edu/courses/ec-716-d-lab-waste-fall-2015/ https://unccelearn.org/course/view.php?id=131&page=overview https://unccelearn.org/course/view.php?id=87&page=overview Industry Mapping:	5	Mapping the supply chain of different waste management	

		Waste & Construction and Demolition Waste. Management of different waste streams – collection, transportation, treatment, storage and disposal. Basel Convention, Extended producer responsibility (EPR) Energy & Resource Recovery - Incineration , Co-processing, Composting , Bio-methanation, Management			gement systems and finding the issues & challenges.	
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		nt of solar photo- voltaic modules or panels or cells, Battery Waste Manageme nt Rules, Circular Economy				
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Textbooks:

This syllabus has been designed for the United Nations Sustainability Development Goals, so there are no prescribed textbooks. Please refer to the study material and online courses.

Course Code- PCCCS481

Course Title – Data Analytics

Credit – 1

Category – Professional Core Course (Sessional)

Semester – IV

L:T:P:S – 0:0:0:2

Modul e No.	Topic	Sub-topics	Mapping with Industry and International Academia	Lectur e	Correspondin g Lab	Textboo k
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				Hours	Assignments	Mapping
1	Introduction to Data Analytics	Definition and Importance of Data Analytics, Types of Data (Structured, Unstructured, Semi-structured), Applications of Data Analytics in Real-world Scenarios (e.g., fraud detection, recommendation systems, precision agriculture) Data Analytics Workflow: Data Collection, Cleaning, Analysis, and Reporting Overview of Data Analytics Tools (Excel, Python,	International Academia: How to Process, Analyze and Visualize Data: https://ocw.mit.edu/courses/res-c-00S-how-to-process-analyze-and-visualize-data-january-iap-2012/ Introduction to R and Geographic Information Systems (GIS) https://ocw.mit.edu/courses/introduction-to-r-and-gis-fall-2023/ Industry Mapping: Power BI, R, Excel	5	1. Basic dataset exploration using tools using Excel. Basic Excel functions like VLOOKUP, HLOOKUP, INDEX etc. Creation of Pivot table. 2. Reading data from text, web, and Excel files and exploring various commands for descriptive analysis. 3. Basic Power BI functions (Aggregation, text, logical, filter, ranking, statistical, data visualization, table manipulation)	

		Tableau, Power BI, and R)			4. Basic data handling using R: importing, exploring, and manipulating datasets	
2	Data Preprocessing and Cleaning	Data Quality Issues and Handling Missing Data in R, Techniques for Data Cleaning (Removing duplicates, fixing invalid entries), Data Transformation and Standardization (Normalization, Encoding), Introduction to Python Libraries for Data Preprocessing (Pandas, NumPy), Handling Outliers and Imbalanced Data	International Academia: How to Process, Analyze and Visualize Data: https://ocw.mit.edu/courses/res-c-00S-how-to-process-analyze-and-visualize-data-january-iap-2012/ Introduction to R and Geographic Information Systems (GIS) https://ocw.mit.edu/courses/introduction-to-r-and-gis-fall-2023/ Industry Mapping: Power BI, R, Excel	5	1. Hands-on exercises on data cleaning and preprocessing using R. 2. Working on datasets with missing and noisy data.	

3	Data Visualization and Exploratory Data Analysis (EDA)	Importance of Data Visualization in Decision-making, Visualization Tools and Libraries (Matplotlib, Seaborn, Tableau Overview), Plotting Techniques: Bar Charts, Line Charts, Pie Charts, Histograms, Scatter Plots, Heatmaps, Basics of Exploratory Data Analysis (EDA), Identifying Trends, Patterns, and Outliers in Data	International Academia: Statistics and Visualization for Data Analysis and Inference: https://ocw.mit.edu/courses/res-S-0002-statistics-and-visualization-for-data-analysis-and-inference-january-iap-2005/ Introduction to R and Geographic Information Systems (GIS) https://ocw.mit.edu/courses/introduction-to-r-and-gis-fall-2023/ Industry Mapping: Power BI, R, Tableau	8	1. Creating visualizations to interpret datasets using Matplotlib and Seaborn. 2. Conducting EDA on sample datasets (e.g., sales data, sensor data). 3. Apply and explore various plotting functions on a dataset. (a) Normal curve (b) Density and contour plot (c) Correlation and scatter plot (d) Histogram (e) Three dimensional plotting	
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					4. Create visualizations: histograms, scatter plots, boxplots, and bar plots using ggplot2 using R and using Power BI.	
4	Basic Statistical and Machine Learning Techniques	Descriptive Statistics (Mean, Median, Mode, Standard Deviation, Variance, Correlation), Basics of Probability in Data Analytics, Introduction to Regression (Linear, Polynomial) and Classification, Basics of Machine Learning Workflow: Training, Testing, and Validation, Evaluating	International Academia: Statistics and Visualization for Data Analysis and Inference: https://ocw.mit.edu/courses/res-S-0002-statistics-and-visualization-for-data-analysis-and-inference-january-iap-2005/ Introduction to R and Geographic Information Systems (GIS) https://ocw.mit.edu/courses/introduction-to-r-and-gis-fall-2023/	6	1. Implement basic statistical functions in Excel. 2. Implementing simple regression models (Linear Regression) using R. 3. Building a basic classification model (e.g., Logistic Regression) in R. 4. (a) Compute mean, median, mode, variance,	

		Model Performance (Accuracy, Precision, Recall, F1-Score, Confusion Matrix), Introduction to Python's Scikit-learn Library. Analyze datasets using basic statistical measures and visualization tools in R.			and standard deviation using R, (b) Analyze distributions and correlations. 5. Creating data analysis dashboard using Power BI.	
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Projects: The projects will be performed by groups of students.
1. Sales and Customer Analysis Dashboard creation using Power BI. 2. Scrape IMDB movie ratings and details using R and save the details of top movies to .csv file. 3. Create a dashboard for COVID-19 Data Analysis using Power BI. Tasks: Import a public COVID-19 dataset with columns like Date, Country, New Cases, New Deaths, and Total Vaccinations, Transform data: Group data by Country or Region, Calculate daily growth rates for cases and deaths. Visualizations: Global Cases and Deaths Trends (Line Chart), Cases by Country (Map Visualization), Vaccination Progress (Clustered Column Chart). Outcome: An interactive dashboard presenting global COVID-19 trends. 4. Exploratory Data Analysis on Iris Dataset using R/Python.

5. Create a simple dashboard to analyze sales performance using Power BI. Import a sample sales dataset with columns like `Order ID`, `Date`, `Sales`, `Profit`, `Category`, and `Region`, perform basic data cleaning in Power Query (e.g., handle null values, format dates), create visualizations as follows:

- Total Sales (Card).
- Sales by Region (Map).
- Monthly Sales Trends (Line Chart).
- Sales by Category (Bar Chart or Pie Chart).

Then add a slicer for filtering by year or region. The outcome will be a basic dashboard showcasing sales performance across regions and categories.

6. House Price Prediction using Machine Learning in Python/R.
7. Create a dashboard for **Employee Performance Analysis using Power BI**. Import an employee dataset with columns like `Employee ID`, `Name`, `Department`, `Salary`, `Performance Score`, and `Joining Date`, Create calculated columns:
- `Years of Service = DATEDIFF(Joining Date, TODAY(), YEAR)`.
 - `Performance Tier = IF(Performance Score > 80, "High", IF(Performance Score > 50, "Medium", "Low"))`.

Visualizations:

- Salary Distribution (Histogram).
- Performance by Department (Clustered Bar Chart).
- Employee Count by Performance Tier (Pie Chart).

It will output a report highlighting employee performance and salary distribution.

8. Zomato Data Analysis using Python/R.
9. Analyze sales performance and trends using Excel operations. **Tasks:** Import a sales dataset with columns like `Order ID`, `Date`, `Product`, `Category`, `Sales Amount`, `Quantity`, and `Region`.
- Calculate Total Sales, Average Sales, and Quantity Sold using basic formulas.
 - Use **Filters** and **Sort** to analyze sales for specific categories or regions.
 - Create a **Pivot Table** to show total sales by product category and region.
 - Visualize:
 - Monthly sales trends (Line Chart).
 - Sales by region (Bar Chart or Pie Chart).
 - Identify the top-performing product and region.
10. Loan Approval Prediction including visualization of data and data preprocessing using Python/R.

11. Explore customer data to understand demographics and spending behaviour using Excel operations. **Tasks:** Dataset includes Customer ID, Name, Age, Gender, Location, Total Spend.

- o Calculate average customer spend and age using basic formulas.
- o Use **COUNTIF** to calculate the number of male vs. female customers.
- o Create a **Pivot Table** to summarize total spend by location.
- o Visualize:
 - § Age distribution (Histogram).
 - § Spending by gender (Bar Chart).
- o Use **Slicers** in Pivot Tables to filter data by gender or location

Uber Rides Data Analysis using Python/R.

Textbooks:

1. “Data Analytics for Beginners” by Aileen Nielsen
2. “Practical Statistics for Data Scientists” by Peter Bruce and Andrew Bruce, O'Reilly Media, Inc.
3. “Data Analytics Using R”, Seema Acharya, Mc Graw Hill.
4. “Fundamentals of Data Visualization”, Claus O. Wilke. O'Reilly Media, Inc.

Online Learning Resources:

- Complete Guide to Power BI for Data Analysts by Microsoft Press: <http://surl.li/gmcctm>
- Excel Data Visualization: <https://shorturl.at/IQwfM>
- Creating Interactive Tableau Dashboards: <https://shorturl.at/HqeeT>
- R for Data Science: Analysis and Visualization: <https://shorturl.at/oUI8H>